

Remarks on the Generalized Singular Value Decomposition: A Methodology for Dependent Data Fusion



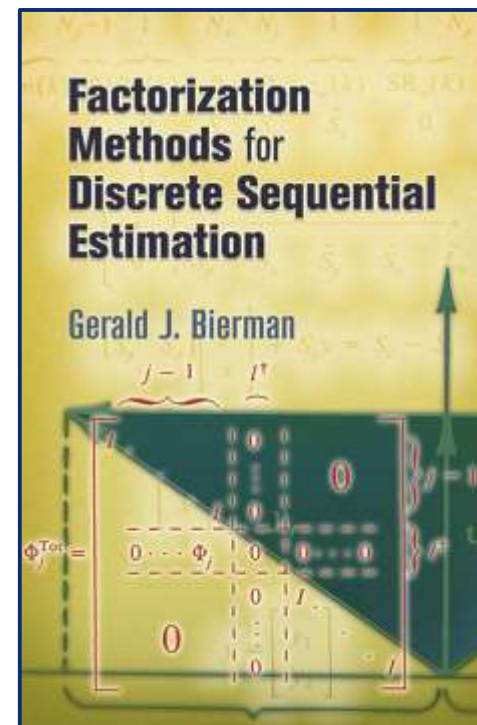
Alexander Michos



Internship Mentor: Dr. Michael Williams

Objectives:

- **Introduction** to Data Fusion, Tracking, SRIF, and Dependent Information.
- Encourage adding the Generalized Singular Value Decomposition to your data-fusion **toolbox**.
- Provide **intuition** for the Gen. Singular Values and **meaning** behind the GSVD Basis.
- Provide **context** for the following presentation by Sofia Taylor.

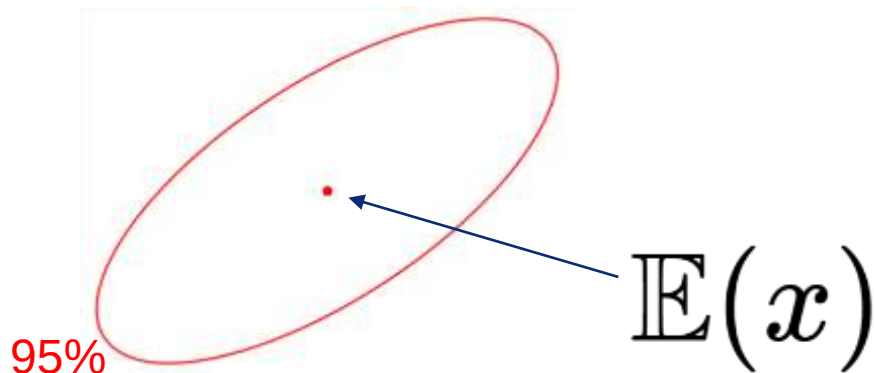


Definitions & Background:

- **Tracks:** A state estimate surrounded by Gaussian uncertainty described by a **Covariance Matrix**.

$$P = \mathbb{E}(xx^{\top})$$

- **Covariance Ellipse:** Spatial confidence interval about the mean / maximum likelihood estimate (**MLE**) at the center.



$$I = P^{-1}$$

Given new updates, we must **fuse/update** our knowledge of the state.

Standard Data Fusion:

- Are given an existing track $P_{(-)}, x_{(-)}$ and receive an incoming **measurement** or **updated estimate** P_{ν}, x_{ν}
 - **Kalman Filters**, or equivalently **Square Root Information Filters (SRIF)**, fuse tracks into a new MLE together with its uncertainty.
 - Solves a **Least-Squares** Problem
- **Information** of the system is additive:

$$I_{(+)} = I_{(-)} + I_{\nu}$$

$$x_{(+)} = [I_{(-)} + I_{\nu}]^{-1} [I_{(-)}x_{(-)} + I_{\nu}x_{\nu}]$$

Square Root Information Filter:

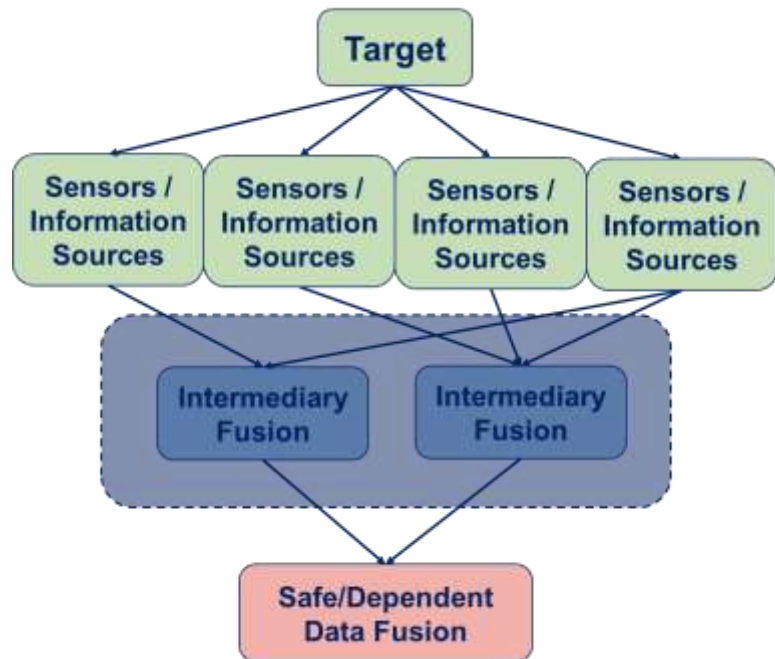
- Information matrices are **symmetric**, **positive-definite**, and so have a family of square roots:
 - Are unique up to **left-rotations** $R \in \sqrt{A} \Leftrightarrow R^\top R = A$
 - Have an upper **triangular** form (Cholesky)
 - QR Decomposition efficiently solves the **Least-Squares update**
- Root-Information space is '**Pythagorean**' and uses **orthogonal rotations**:

$$I_{(+)} = R_{(+)}^\top R_{(+)} = R_{(-)}^\top R_{(-)} + R_\nu^\top R_\nu$$

$$\text{qr} \begin{bmatrix} R_{(-)} & R_{(-)}x_{(-)} \\ R_\nu & R_\nu x_\nu \end{bmatrix} = \begin{bmatrix} R_{(+)} & R_{(+)}x_{(+)} \\ \mathbf{0} & \varepsilon^2 \end{bmatrix}$$

Dependent Fusion Motivation:

- What if measurements are **not independent**? Partially Correlated? Completely Duplicated?



Multi-sensor tracking architectures may pass data into a **complex or unknown pedigree of fusers**.

Duplicate or already **correlated** updates may be erroneously ‘**mixed**’ with independent data.

Dependence in **common** propagation models / times too.

“Unmixing Paint”

Dependent Fusion Implementation:

- If information contained in the two tracks are **correlated**, we must **subtract** the redundancy:

$$I_{(+)} = I_{(-)} + I_{\nu} - I_{(-) \cap \nu}$$

- **Covariance Intersection (CI)**, **Inverse Covariance Intersection (ICI)**, **SRIF**, etc. all have different interpretations on what the **common information** is.
- [Sofia Taylor @ 10:00] **GEM** (Generalized Eigenvalue Method) extracts the **maximal amount of information** from the intersection.
 - What does information intersection mean?

Generalized Singular Value Decomposition:

- Given two matrices, their GSVD produces an SVD-like factorization but with one **orthogonal** component and a **mutual basis of shared information**

Diagonal Weights ($\Sigma_1^2 + \Sigma_2^2 = I$)

$$\begin{array}{rcl} R_1 & = & U_1 \overbrace{\Sigma_1}^{\text{Diagonal Weights}} V \\ R_2 & = & \underbrace{U_2}_{\text{Orthogonal Rotations}} \underbrace{\Sigma_2}_{\text{Diagonal Weights}} \underbrace{V}_{\text{Shared Basis of Information/Content}} \end{array}$$

- The GSVD of a invertible matrix A and B with the same # columns is **equivalent to the SVD** of BA^{-1} , and asymptotically equal in **complexity**.

GSVD as the Basis of Fusion:

- Kalman, Covariance Intersection (CI), Inverse Covariance Intersection (ICI), GEM, etc. are all **operators on the singular values** of the SRIF matrices.

GSVD-SRIF:

$$\begin{aligned} R_{(+)} &= \sqrt{R_{(-)}^{\top} R_{(-)} + R_{\nu}^{\top} R_{\nu}} \\ &= \sqrt{V^{\top} \Sigma_1^2 V + V^{\top} \Sigma_2^2 V} \\ &= \sqrt{V^{\top} V} \\ &\Rightarrow R_{(+)} \sim V \end{aligned}$$

GSVD-ICI:

$$R_{\text{ICI}} = \sqrt{I - (\omega \Sigma_1^{-2} + (1 - \omega) \Sigma_2^{-2})^{-1}} V$$

GSVD-CI:

$$\begin{aligned} I_{(+)} &= \omega I_{(-)} + (1 - \omega) I_{\nu} \\ &= V^{\top} [\omega \Sigma_1^2 + (1 - \omega) \Sigma_2^2] V \\ \Rightarrow R_{(+)} &= \sqrt{\omega \Sigma_1^2 + (1 - \omega) \Sigma_2^2} V \end{aligned}$$

GSVD-GEM:

$$R_{\text{GEM}} = \max(\Sigma_1, \Sigma_2) V$$

GSVD as the Basis of Fusion (Cont):

- Manipulating diagonals is fast **computationally**, helpful in **proofs**, and prime for **intuition**.
 - Computing a GSVD can be done in place of computing the QR.
- **Comparisons:** the **larger** the diagonal, the greater the final **information**, the **tighter** the confidence of the fused result.

Most Confident $\geq$ SRIF $\geq$ GEM $\geq$ CI $\geq$ Least Confident


Greatest
Assumption of
Independence

$$1 \geq \max(\sigma_1, \sigma_2) \geq \sqrt{\omega\sigma_1^2 + (1 - \omega)\sigma_2^2}$$


Least
Assumption of
Independence

GSVD as Universal Fusion:

- Suppose the **common, duplicated** information between two tracks lies in their GSVD basis.
 - Then the resulting fusion can be fully **parameterized** by V and by diagonal weight matrices $\alpha = \text{diag}(\alpha_1, \alpha_2, \dots, \alpha_n)$ $\beta = \text{diag}(\beta_1, \beta_2, \dots, \beta_n)$

$$\text{qr} \begin{bmatrix} \alpha V & \alpha V \vec{x}_1 \\ \beta V & \beta V \vec{x}_1 \end{bmatrix} \longrightarrow x_{(+)} = V^{-1} \frac{\alpha^2 V x_{(-)} + \beta^2 V x_{\nu}}{\alpha^2 + \beta^2}$$

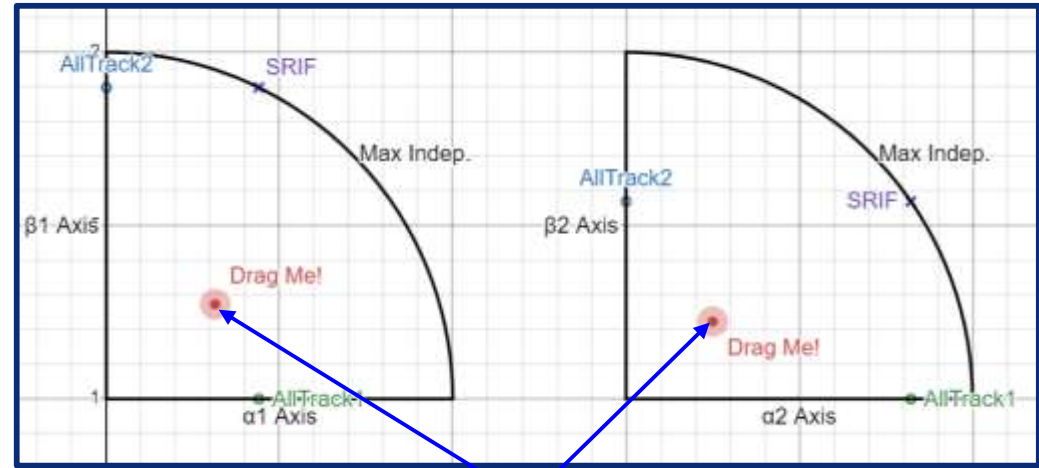
The $\alpha\beta$ magnitude scales the uncertainty of the result.

$$R_{(+)} = \sqrt{V^{\top} \alpha^2 V + V^{\top} \beta^2 V} \sim \sqrt{\overbrace{\alpha^2 + \beta^2}^{\text{uncertainty}}} V$$

Hands-On Demonstration:

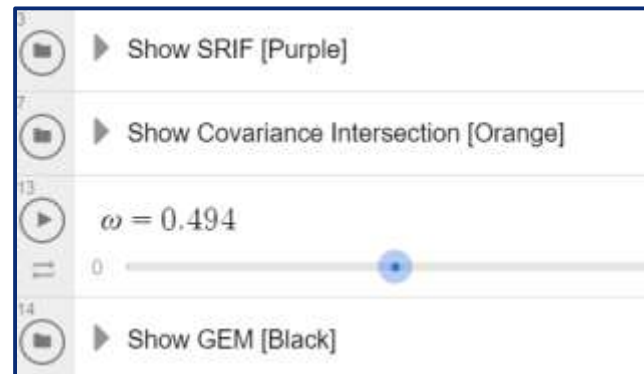
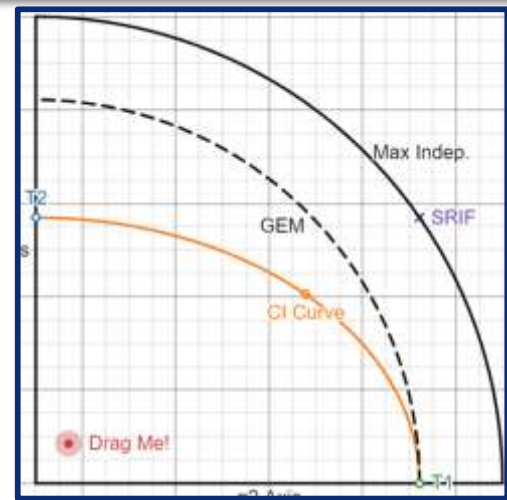
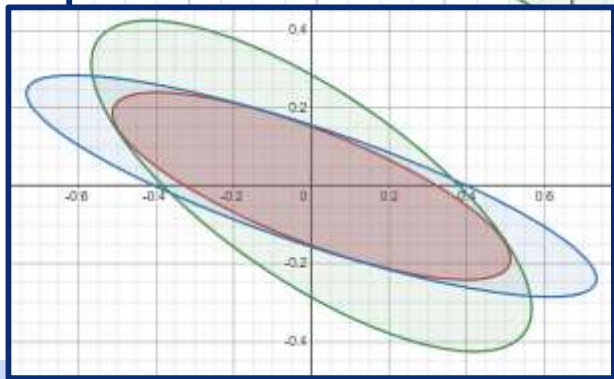
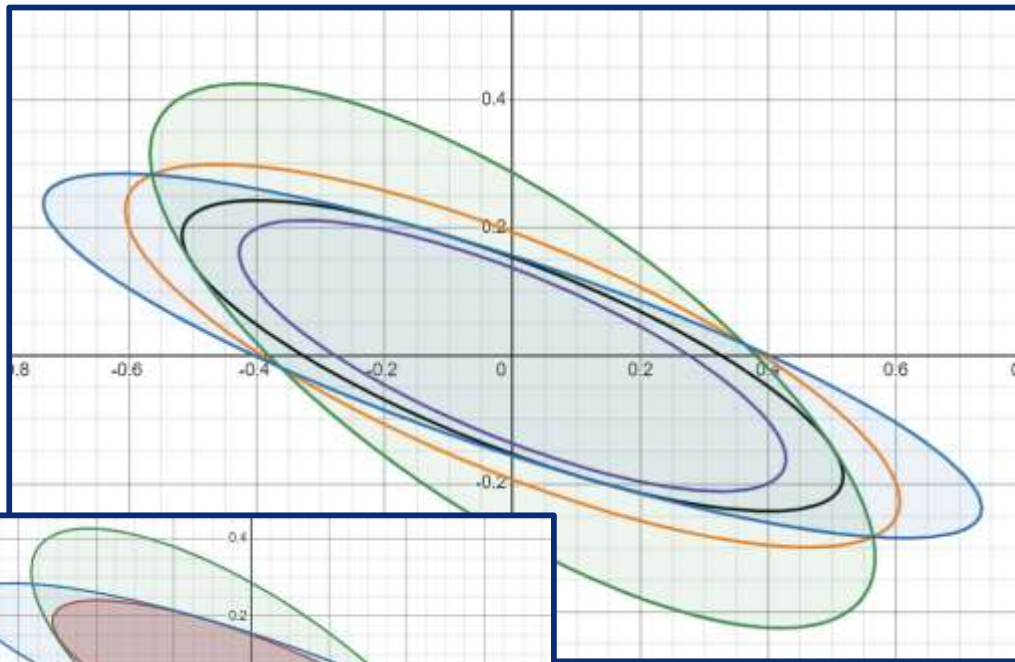


- Let's explore how the $\alpha\beta$ weighting controls and interpolates between **methods of fusion**!



For each of the two singular values, there are $\alpha\beta$ parameters to vary.

Hands-On Demonstration:



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Log-Likelihood Optimization:

- The Log-Likelihood of a **particular track-to-track fusion** or measurement update can be described in Kalman/SRIF terms:
 - How **likely** is the **association**?

$$\mathcal{L} = -\ln \mathbb{P}(x_\nu, P_\nu | x_{(-)} P_{(-)}) = \frac{1}{2} |\varepsilon|^2 + \ln \left(\frac{\det R_{(+)}}{\det R_{(-)} \det R_\nu} \right)$$

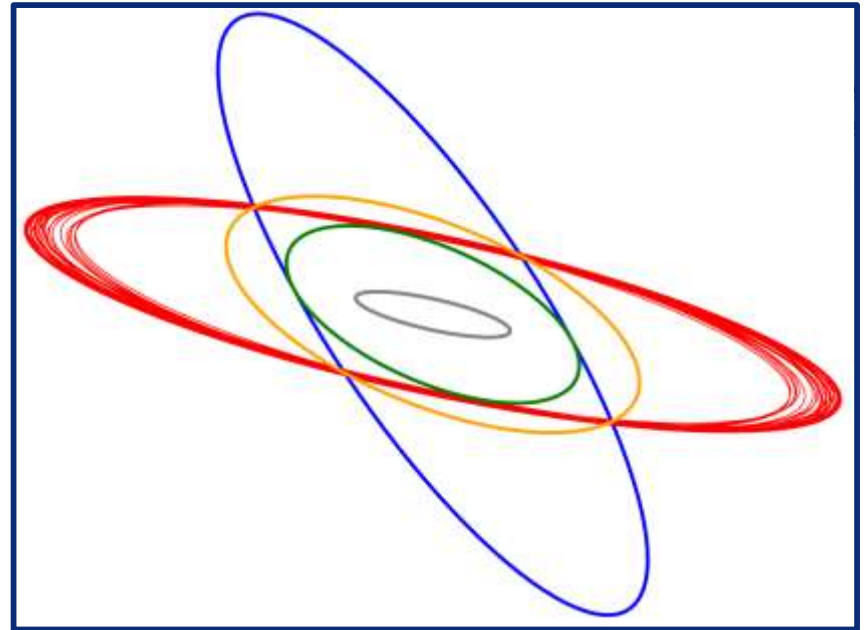
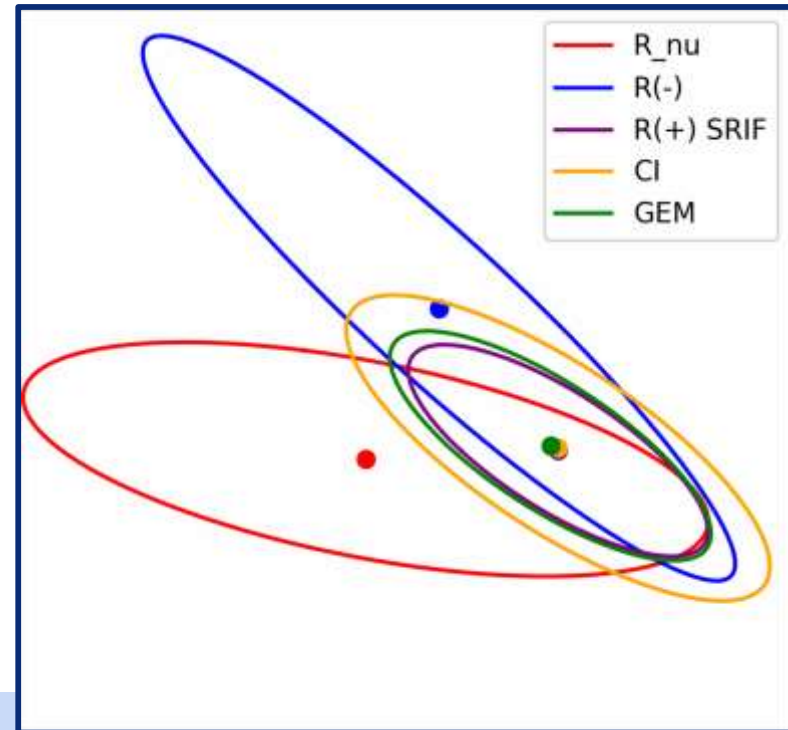
- For the GSVD- $\alpha\beta$ fusion family this reduces to:

$$\mathcal{L} = -\ln(V) + \frac{1}{2} \sum_k \frac{\alpha_k^2 \beta_k^2}{\alpha_k^2 + \beta_k^2} (u_{(-)k} - u_{\nu k})^2 + \ln \left(\frac{\alpha_k^2 \beta_k^2}{\alpha_k^2 + \beta_k^2} \right)$$

- Optimizing over $\alpha\beta$ gives the most **plausible** means of **associating dependent data**.
 - Diagonals are nice: elementary calculus.

Visualizations & The End Result:

- What to do with **non-concentric means**?
 - Define **Non-Concentric GEM** by **likelihood-optimization** along the **arc** in the $\alpha\beta$ plane defined by dependence-radius set by GEM.



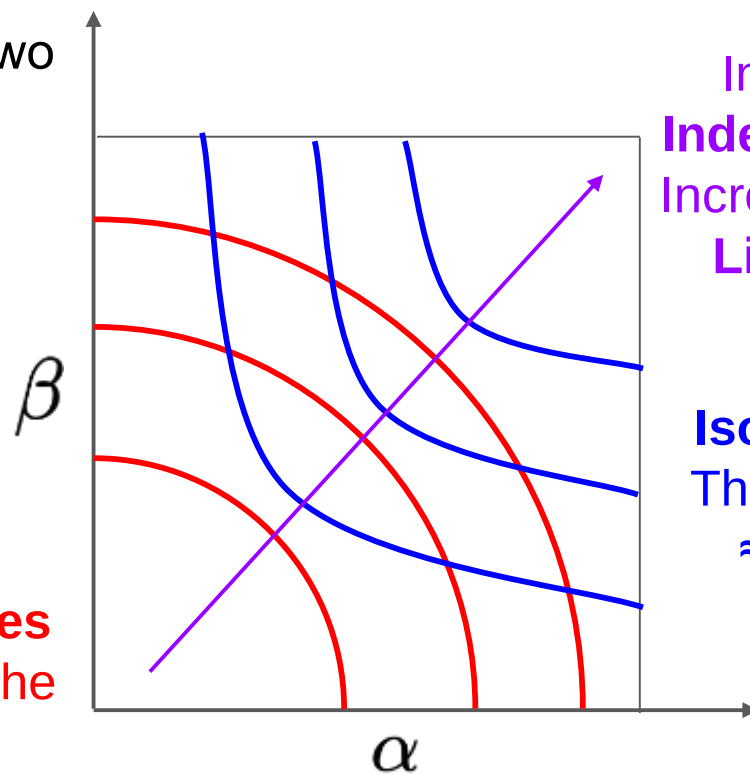
Summary: Nature of Singular Value Space

In $\alpha\beta$ space, there are two important **surfaces**:
Dependence & Likelihood

$$\sqrt{\alpha^2 + \beta^2}$$

Iso-Dependence Circles

The greater the radius the greater the assumed independence of the tracks.



Increasing
Independence,
Increasing **Log-Likelihood**

Iso-Likelihood Hyperbolas

The greater the dependence $\approx$ the less the likelihood.

$$\frac{\alpha^2 \beta^2}{\alpha^2 + \beta^2}$$

Information Inequalities:

GEM provides a fusion when **maximally dependent**

$$\max (\sigma_1, \sigma_2)^2 \leq \alpha^2 + \beta^2 \leq 1$$

SRIF provides a fusion when **maximally independent**

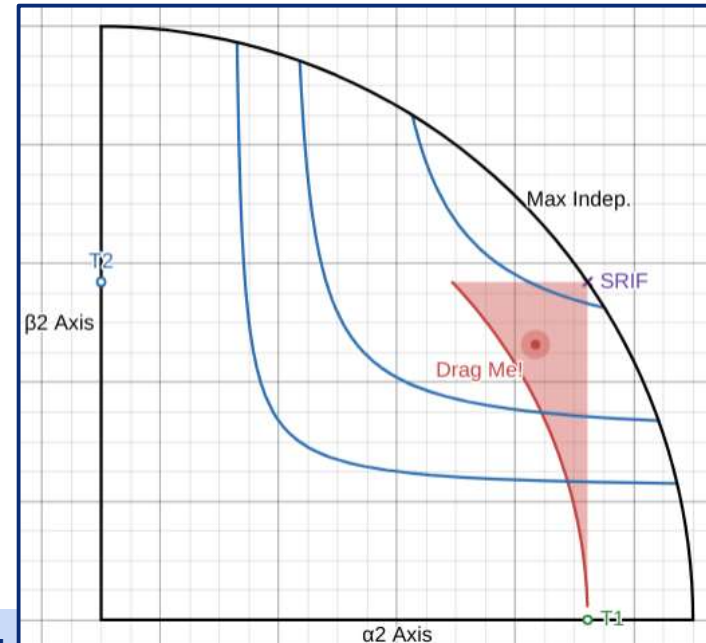
Information in the **new** singular values cannot **exceed** the information in the **original** singular values

$$0 \leq \alpha \leq \sigma_1$$

$$0 \leq \beta \leq \sigma_2$$

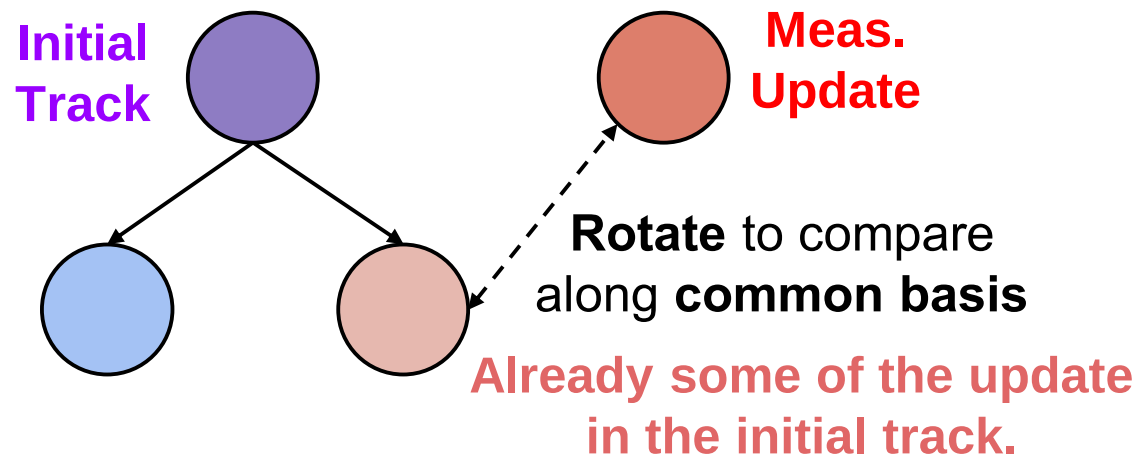
Show Iso-Likelihood & Iso-Dependence Contours

$$L_{\text{ogLikelihood}} = \frac{1}{2} \ln \left(\frac{\alpha_1^2 \beta_1^2}{\alpha_1^2 + \beta_1^2} \frac{\alpha_2^2 \beta_2^2}{\alpha_2^2 + \beta_2^2} \right)$$



Deeper GSVD Basis Intuition:

- GEM includes a row-update if it **exceeds the content** of the update already existent in the track.
 - Maximally removing the existing **content** gives internally **tangent ellipses**
- **Any** orthogonal rotation which **aligns rows** provides a measure of the content:
 - Once aligned, **compare the magnitudes** $\Leftrightarrow$ **compare singular values**



$$Q_1 R_1 = \begin{bmatrix} \lambda_1 v \\ \vdots \end{bmatrix} \quad \lambda_1 \geq \lambda_2?$$
$$Q_2 R_2 = \begin{bmatrix} \lambda_2 v \\ \vdots \end{bmatrix}$$

Questions?

Thank You for Listening!



Always More to Talk About:

- The Bar Shalom-Campo Formula gives an exact solution to the fusion of two tracks when their **cross-covariance is known**.
 - If the cross-covariance lies in the GSVD basis, we can describe a 'dependent-data-fusion' method **in terms of BSC**.
- The **content of a unit vector in a SRIF matrix** (hence the nonzero gen. sing. value) is related to the distance along the vector to the information ellipse.
- Rotating and then taking the $\text{Max}(\sigma_1, \sigma_2)$ or $\text{max}(\lambda_1, \lambda_2)$ operator is the **tangency operator**: giving an axis where the fused ellipse touches internally.
- The GSVD basis is the **basis of independence**: a set of vector updates, once put in the basis, **commute** under the GEM update scheme.
- **Dependence in Time/Propagation** & GSVD-based methods of propagation.